

VHC COMMUNICATION BASIC SPECIFICATION

Version	Document			Model
1.0.0	VHC COMMUNICATION BASIC SPECIFICATION			
Date	Drawn	Checked	Approved	Page

REVISION HISTORY

VER	DATE	DESCRIPTION	Note
2.3	2018-03-07	Cassette Type added to S2F49 and S6F11(RPTID = 4). Port Type added to Current Port States of S1F4. Scan Command Scenario added. Carrier Type VID added. RPTID 16 added. CEID 153,154(VehicleCarrierInstalled/Removed) added.	
2.5	2018-03-10	SCAN Command changed. CarrierLoc deleted from CarrierInstalled/Removed event. CarrierLoc added to RPTID 11.	
2.6	2018-03-28	Cassette Position added.	
3.0	2018-09-10	Manual Port added	
3.1	2018-09-18	CEID 158, 159 / RPTID 33, 34 Added	

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1. INTRODUCTION

1.1. Purpose

The purpose of this document is to define the communication specifications to be used in online-communication with the intrabay/interbay system..

1.2. Referenced Documentation

No	SEMI No.	Title
1	SEMI E5	SEMI Equipment Communications Standard 2 Message Content (SECS-II)
2	SEMI E30	Generic Model For Communications and Control of Manufacturing Equipment (GEM)
3	SEMI E37	High-Speed SECS Message Services (HSMS) Generic Services
4	SEMI E37.1	High-Speed SECS Message Services Single-Session Mode (HSMS-SS)
5	SEMI E82	Specification for INTERBAY/INTRABAY AMHS SEM (IBSEM)

2. TERMINOLOGY

Term	Explanation
AGV (MGV)	Automatic (manual), the device's Vehicle
OHS	High-Shuttle
MCT	Mask Cassette Transfer
Loader Port	Infer-face position of equipment where cassette is delivered
Transfer Port	When Cassette is the emergence, on the point of the transfer system cassette ownership is changed
Cassette	Container glass is put
Cassette ID	ID cassette is separated by

3. BASIC COMMUNICATION STANDARD

3.1. Communication Protocol

- Communication : TCP/IP
- Communication Speed : 10/100Mbps

3.2. Overview of HSMS Message

Byte	Description
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0~3 Bytes	Message length (Notify message header and bytes of message text)
4~13 Bytes	Message header
14~n Bytes	Message text (Format notified on PType of message header)

3.3. Message Format

Length (0~3 bytes)	Header (4~13 bytes)	Text (14~n bytes)
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- Message Length Field
 - The Format of the message Length is 4 byte unsigned integer, writes bytes of message header and message text
- Message Header Field

Bytes	Description
0~1	Session ID (Device ID)
2	Header Byte 2
3	Header Byte 3
4	PType
5	SType
6~9	System Bytes

※ PType, SType = 0, The message text includes a validSECS-II message

3.4. Communication Standard Summary

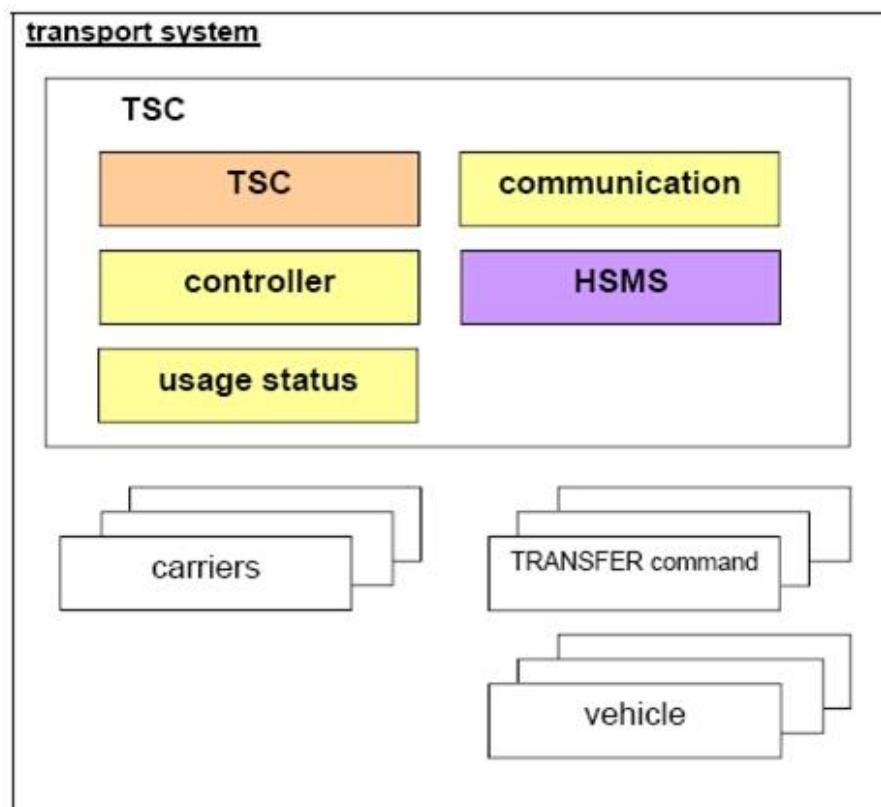
No.	Symbol	Parameter Name	Description	Default	Range	Resolution
1	N.A.	Host Name	MCS Name	TBD	-	N.A.
2	N.A.	TCP Port Number	TCP Port Number	TBD	1-65535	
3	N.A.	MCS Side IP Address	MCS IP Address	TBD	-	-
4		TSC Side IP Address	TSC IP Address	TBD	-	-
5	N.A.	Connection Mode	Connection Mode	Passive for TSC	Passive only for Eqp. side	N.A.
6	T3	Reply TimeOut	Reply message timeout occur	45 sec	1-120 sec	1 sec
7	T5	Connection Separation TimeOut	Trial time interval when Connection is failed	10 sec	1-240 sec	1 sec
8	T6	Control TimeOut	Control message timeout occur	5 sec	1-240 sec	1 sec
9	T7	Connection Idle TimeOut	Before connection failure occurred, The time until first connection is succeed for using to HSMS communication	10 sec	1-240 sec	1 sec
10	T8	Network Character TimeOut	In a message, timeout happened between the connection characters	5 sec	1-120 sec	1 sec
11	N.A.	Maximum Message Size	When Communication, The maximum Message Size to send or received	4096	-	-
12	N.A.	Maximum Concurrent Opened Transactions	After access to HSMS The maximum number the concurrent possible transactions	1024	-	-
13	N.A.	Number of Device ID to be used	Number of max equipment to be connected to HSMS	9	-	-

※ TBD = To be defined

4. IBSEM STATES MODEL

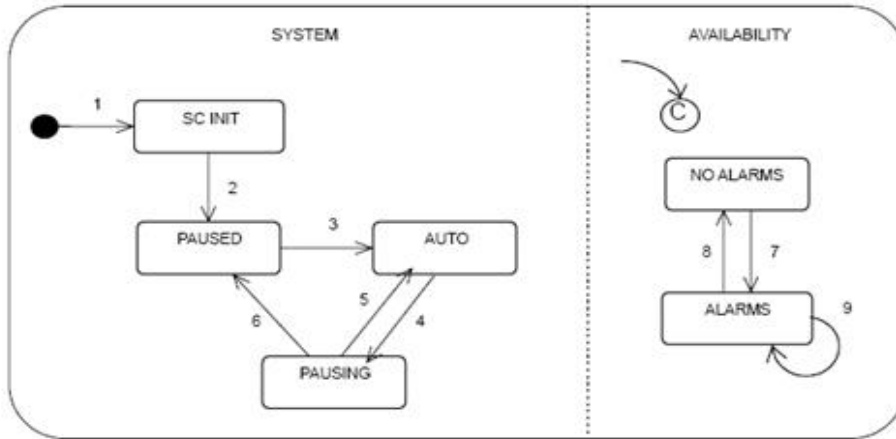
4.1. TSC State Model Entities

The following illustration indicates the entities contained in the state models listed on the previous page. For example, in the TSC State Model, the TSC is the entity in that model. Likewise, in the case of the Carrier State Model, the carrier is the entity in that model. So, in the TSC State Model, there is only one entity, the TSC. However, in the Carrier State Model, there are as many entities as there are carriers. The following illustration indicates how things look from the host's perspective but the standard does not define whether or not this is the actual condition.



4.2. TSC State Model

4.2.1. TSC State Model



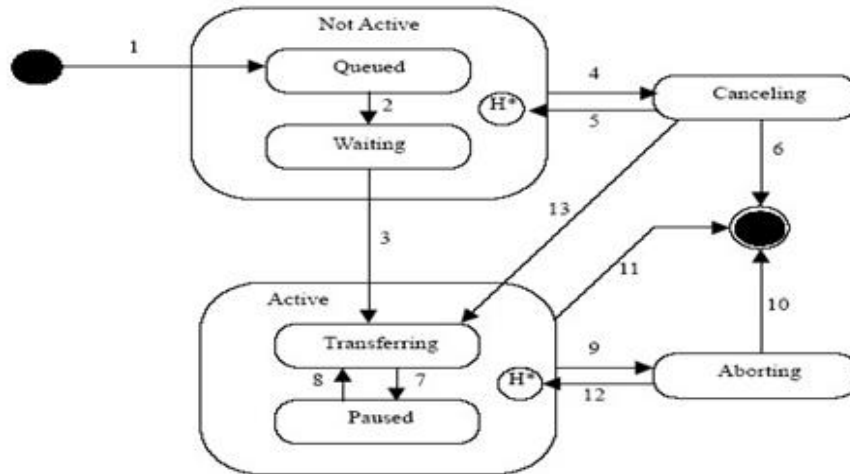
4.2.2. TSC State Transition Table

No	Previous State	Trigger	New State	Actions	Comments
1	NONE	TSC Initiation	TSC INIT	S6F11 TSCAutoInitiated	System runs through its startup sequence
2	TSC INIT	System started up successfully. All Cassette movement stopped	PAUSED	S6F11 TSCPaused	System ready
3	PAUSED	RESUME command	AUTO	S6F11 TSCAutoCompleted	System will now execute remote commands.
4	AUTO	PAUSE command	PAUSING	S6F11 TSCPauseInitiated	Carriers that are currently moving may continue to move but they must not begin another movement.
5	PAUSING	RESUME command	AUTO	S6F11 TSCAutoCompleted	System will now execute remote commands.
6	PAUSING	All Cassette movement has competed	PAUSED	S6F11 TSCPauseCompleted	System will accept and queue new commands but will not execute them.
7	NO ALARMS	Alarm Set	ALARMS	S6F11 AlarmSet	TSC unit error is happened
8	ALARMS	All Alarms cleared	NO ALARMS	S6F11 AlarmCleared	TSC unit error is cleared

9	ALARMS	Alarm Set	ALARMS	S6F11 AlarmSet	Alarm occurs when there is already an outstanding alarm.
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4.3. Transfer Command State Model

4.3.1. Transfer Command State Model



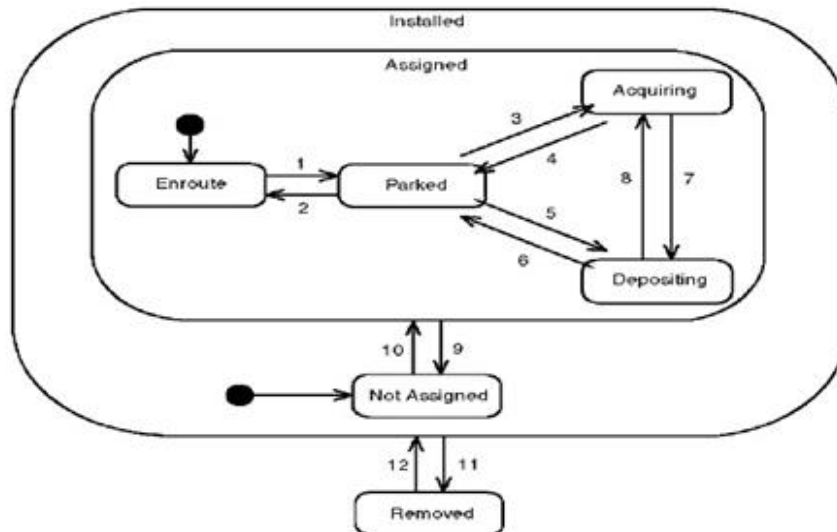
4.3.2. Transfer Command State Transition Table

No	Previous State	Trigger	New State	Actions	Comments
1	NONE	The Host generated TRANSFER command is successfully acknowledge by the TSC	QUEUED		
2	QUEUED	The TRANSFER command has been initiated by the TSC	WAITING	S6F11 TransferInitiated	
3	WAITING	The acquire of the first carrier of the transfer unit begins	TRANSFERRING	S6F11 Transferring	
4	NOT ACTIVE	Host sends CANCEL command for a specified TRANSFER command to TSC	CANCELING	S6F11 TransferCancelInitiated	
5	CANCELING	Transport system is unable to cancel the TRANSFER command	Previous NOT ACTIVE sub-state	S6F11 TransferCancelFailed	

No	Previous State	Trigger	New State	Actions	Comments
6	CANCELING	The cancel procedure for the TRANSFER command has completed by the transport system and TSC	NONE	S6F11 TransferCancelCompleted	
7	TRANSFERRING	The TSC resumes execution of the TRANSFER command due to an anomaly condition	PAUSED	S6F11 TransferPaused	
8	PAUSED	The TSC resumes execution of the TRANSFER command since the anomaly condition has been cleared	TRANSFERRING	S6F11 TransferResumed	
9	ACTIVE	Host initiates an abort of TRANSFER command	ABORTING	S6F11 TransferAbortInitiated	
10	ABORTING	The abort procedure for the TRANSFER command has completed by the transport system and TSC	NONE	S6F11 TransferAbortCompleted	
11	ACTIVE	The TRANSFER command has completed by the transport system and SC(either successfully or unsuccessfully)	NONE	S6F11 TransferCompleted	ResultCode ResultCode=0 : success nonzero : unsuccess
12	ABORTING	Transport system is physically unable to abort the TRANSFER command	Previous ACTIVE substate	S6F11 TransferAbortfailed	
13	CANCELING	Transport system is unable to cancel because the transfer is now ACTIVE	TRANSFERRING	S6F11 Transferring	

4.4. Vehicle State Model

4.4.1. Vehicle State Model



4.4.2. Vehicle State Transition Table

No	Previous State	Trigger	New State	Actions	Comments
1	ENROUTE	Vehicle arrives at a transfer port associated with an ACTIVE transfer command	PARKED	S6F11 VehicleArrived	
2	PARKED	Vehicle departs a transfer port associated with an ACTIVE transfer command	ENROUTE	S6F11 VehicleDeparted	
3	PARKED	The carrier handoff parallel I/O starts for the vehicle to acquire (load) the transfer unit	ACQUIRING	S6F11 VehicleAcquireStarted	
4	ACQUIRING	The carrier handoff parallel I/O completes for the vehicle to acquire (unload) the transfer unit	PARKED	S6F11 VehicleAcquireCompleted	
5	PARKED	The carrier handoff parallel I/O starts for the vehicle deposit (unload) the transfer unit.	DEPOSITING	S6F11 VehicleDepositStarted	

No	Previous State	Trigger	New State	Actions	Comments
6	DEPOSITING	The carrier handoff parallel I/O completes for the vehicle to deposit (unload) the transfer unit	PARKED	S6F11 VehicleDepositCompleted	
7	ACQUIRING	The carrier handoff parallel I/O completes for the vehicle to acquire (load) the transfer unit and starts for the vehicle to deposit (unload) the carrier	DEPOSITING	S6F11 VehicleDepositStarted	
8	DEPOSITING	The carrier handoff parallel I/O completes for the vehicle to deposit (unload) the transfer unit and starts for the vehicle to acquire (load) the carrier	ACQUIRING	S6F11 VehicleAcquireStarted	
9	ASSIGNED	Vehicle is no longer being utilized for the specified command	NOT ASSIGNED	S6F11 VehicleUnassigned	
10	NOT ASSIGNED	Vehicle is allocated to TRANSFER command	ASSIGNED	S6F11 VehicleAssigned	
11	INSTALLED	Vehicle is removed from use of transfer commands	REMOVED	S6F11 VehicleRemoved	
12	REMOVED	Vehicle is installed for use of transfer commands	INSTALLED	S6F11 VehicleInstalled	

4.5. Carrier State Model

4.5.1. Carrier State Model

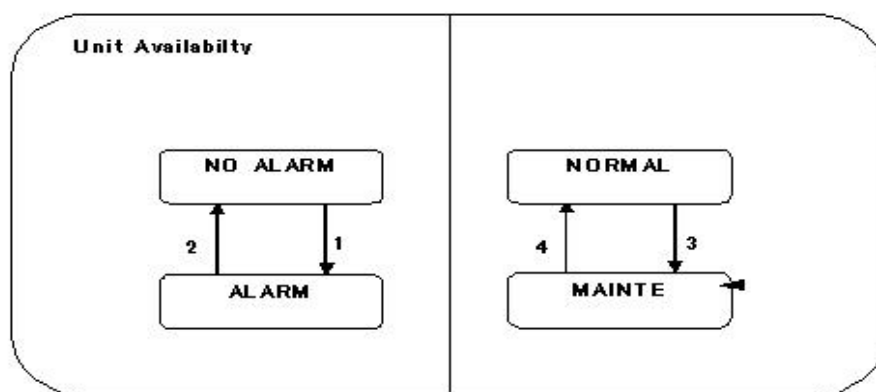


4.5.2. Carrier State Transition Table

No	Previous State	Trigger	New State	Actions	Comments
1	NONE	A TSC database entry is created for the carrier	INSTALLED	S6F11 CarrierInstalled	
2	INSTALLED	A TSC database entry is removed for the carrier	NONE	S6F11 CarrierRemoved	

4.6. Unit Alarm State Model

4.6.1. Unit Alarm State Model



4.6.2. Unit Alarm State Transition Table

No	Previous State	Trigger	New State	Actions	Comments
1	NO ALARM	Alarm occurred	ALARM	S6F11 UnitAlarmSet	
2	ALARM	Alarm Cleared	NO ALARM	S6F11 UnitAlarmCleared	
3	NORMAL	Maintenance	MAINTE	S6F11 UnitAlarmSet	
4	MAINTE	No Maintenance	NORMAL	S6F11 UnitAlarmCleared	

4.7. Vehicle Blocking State Model

4.7.1. Vehicle Blocking State Model

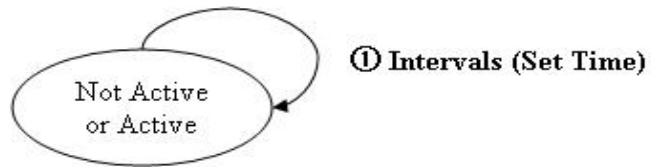


4.7.1. Vehicle Blocking State Transition Table

No	Previous State	Trigger	New State	Actions	Comments
1	NOT BLOCKED	Vehicle Moving is Blocking	BLOCKING	S6F11 VehicleBlockingStarted	
2	BLOCKING	Vehicle Moving is Resumed	NOT BLOCKED	S6F11 VehicleBlockingEnded	

4.8. Monitored Vehicles State Model

4.8.1. Monitored Vehicles State Model

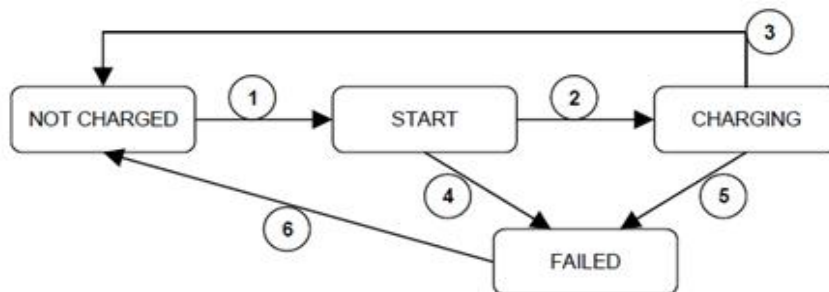


4.8.2. Monitored Vehicles State Transition Table

No	Previous State	Trigger	New State	Actions	Comments
1	NOT ACTIVE ACTIVE	Intervals (Set Time)	NOT ACTIVE ACTIVE	S6F11 MonitoredVehicles	

4.9. Vehicle Charge State Model [AGV ONLY]

4.9.1. Vehicle Charge State Model



4.9.2. Vehicle Charge State Transition Table

No	Previous State	Trigger	New State	Actions	Comments
1	NOT CHARGED	Vehicle Moving Starts to Charge Station	START	S6F11 VehicleChargeSTStarted	
2	START	Vehicle Charging Starts in Charge Station	CHARGING	S6F11 VehicleCharging	
3	CHARGING	Vehicle Charging Completes in Charge Station	NOT CHARGED	S6F11 VehicleNotCharged	
4	START	Status of Vehicle is Not AUTO	FAILED	S6F11 VehicleChargeFailed	
5	CHARGING	Status of Vehicle is Not AUTO	FAILED	S6F11 VehicleChargeFailed	

6	FAILED	Status of Vehicle is AUTO	NOT CHARGED	S6F11 VehicleNotCharged	
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4.10. Carrier State Model (Manual Port)

4.10.1. Carrier State Transition Table (Manual Port Only)

<i>No</i>	<i>Previous State</i>	<i>Trigger</i>	<i>New State</i>	<i>Actions</i>	<i>Comments</i>
1	WAIT IN	Carrier has arrived at the input port.	WAIT IN	S6F11 CarrierWaitIn	INPUT Port
2	WAIT OUT	Carrier has arrived at the outputport..	WAIT OUT	S6F11 CarrierWaitOut	OUTUT PORT

5. DATA DICTIONARY ITEM

5.1. SECS Data Items

No.	Name	Format	Size	Explanation
1	ACKC5	B	1	Acknowledge Code -. 0 = ACK -. >0 = Error, Could not ACK -. 1~63 = Hold
2	ACKC6	B	1	Acknowledge Code -. 0 = ACK -. >0 = Error, Could not ACK -. 1~63 = Hold
3	ACKC10	B	1	Acknowledge Code -. 0 = ACK -. 1 = Message not displayed -. 2~63 = Hold
4	ALCD	B	1	Acknowledge Code -. bit8 = 1 Generated alarm status -. bit8 = 0 Clear alarm status -. bit 1~7 = 0 not used
5	ALED	B	1	Alarm Enable/Disable code -. bit8 = 1 Alarm enabled -. bit8 = 0 Alarm disabled
6	ALID	U4	4	Alarm ID
7	ALTX	A	Max:80	Alarm Text. (Max 80 Char)
8	CEED	BOOLEAN	1	Collection event or trace enable/disable code -. false = Invalid -. true = Valid
9	CEID	U2	2	Collection event ID
10	CEPACK	L, U1	1	Command Enhanced Parameter Acknowledge -. 0 = No Error -. 1 = CPNAME doesn't exist -. 2 = the incorrect value is specified in CEPVAL -. 3 = the incorrect format is specified in CEPVAL -. 4 = the CPNAME usage isn't valid
11	CEPVAL	A, B, BOOLEAN, Fx, Ix, J, L, Ux	40	Command Enhanced Parameter Value
12	COMMACK	B	1	Establish Command Acknowledge Code -. 0 = ACK -. 1 = Denied, Retry -. 2~63 = Hold

No.	Name	Format	Size	Explanation
13	CPACK	B	1	Command Parameter Acknowledge Code -. 0 = No Error -. 1 = Undefined Parameter -. 2 = the incorrect value is specified in CPVAL -. 3 = the incorrect format is specified in CPVAL -. 4~63 = Reserved
14	CPNAME	A	Max:40	Command Parameter Name -. 'DESTLOC' : Destination shelf Location or port ID (10101;bank-bay-level, A2STK01S_TO01) -. 'LOCTYPE' " Location Type ('SHELF', 'PORT')
15	CPVAL	A, B, BOOLEAN, Fx, Ix, K, I, Ux	40	Command Parameter Value
16	DATAID	U4	4	Data ID -. The emulator always sends DATAID = 0
17	DRACK	B	1	Define Report Acknowledge -. 0 = ACK -. 1 = Denied, insufficient space -. 2 = Denied, incorrect format -. 3 = Denied, at least one RPTID is already defined -. 4 = Denied, at least VID is already defined -. 5~63 = Reserved
18	EAC	B	1	Equipment Acknowledge Code -. 0 = ACK -. 1 = Denied, there isn't at least one constant -. 2 = Denied, busy -. 3 = Denied, at least constant is beyond the allowed range -. >3 = Some other error -. 4~63 = Hold
19	ECDEF	A, B, BOOLEAN, Fx, Ix, K, L, Ux	-	Equipment Constant Default Value
20	ECID	U2	2	Equipment Constant ID
21	ECMAX	A, B, BOOLEAN, Fx, Ix, K, L, Ux	-	Equipment Constant Maximum Value
22	ECMIN	A, B, BOOLEAN, Fx, Ix, K, L, Ux	-	Equipment Constant Minimum Value
23	ECNAME	A	Max:64	Equipment Constant Name
24	ECV	A, B, BOOLEAN, Fx, Ix, K, L, Ux	-	Equipment Constant Value

No.	Name	Format	Size	Explanation
25	ERACK	B	1	Enable/Disable event report acknowledge code -. 0 = ACK -. 1 = Denied, there isn't at least one CEID -. 2~63 = Reserved
26	HCACK	B	1	Host Command Parameter Acknowledge code -. 0 = Confirmed, the command was executed (The transport system doesn't use this value. Confirmation is made using the number 4 value.) -. 1 = Command doesn't exist -. 2 = Currently not able to execute -. 3 = At least one parameter isn't valid -. 4 = Confirmed, the command will be executed and completion will be notified by an event -. 5 = Rejected, already requested -. 6 = Object doesn't exist -. 7 = Already Auto (Only HostCommand(RESUME)) -. 8 = Exceeded the command limit to DEST -. 9 = Exceeded the command limit to SOURCE (8,9 : by Vehicle Controller) -. 13 = Mismatch between CommandID and CarreirID -. 22 = Source Unit can't be found -. 27 = Dest Unit can't be found -. 28 = Source Unit Dest Unit Same -. 29 = Machine Internal Problem -. 31 = Carrier can't be found -. 41 = Source Unit Unavail -. 42 = Dest Unit Unavail -. 43 = Transfer Machine Group does not exist. -. 44 = Transfer Machine does not exist. -. 45 = Source Port Cassette does not exist. -. 46 = DestPort Cassette exist.- -. 47 = SourcePort Cassette Type Interlock -. 48 = Dest Port Cassette Type Interlock -. 49 = TransferMachine CassetteSize Impossibility -. 50 = Machine Call Impossibility -. 55 = CarrierID, already requested
27	LRACK	B	1	Link Report Acknowledge Code -. 0 = ACK -. 1 = Denied, insufficient space -. 2 = Denied, incorrect format -. 3 = Denied, already defined -. 4 = Denied, there isn't at least one CEID -. 5 = Denied, there isn't at least one RPTID -. 6-63 = Reserved
28	MCMD	U2	2	Current SC Mode -. 0 = OFF-LINE MODE -. 1 = ON-LINE REMOTE -. 2 = ON-LINE LOCAL
29	MDLN	A	Max:16	Equipment format
30	MEXP	A	-	Message expected in the form SxxFyy where x is stream and y is function
31	MHEAD	B	10	SECS message block header associated with message block in error

32	OBJSPEC	A	40	The text string used to include the specified object instance point to a specific object instance -. The emulator always sends OBJSPEC = NULL
33	OFLACK	B	1	Acknowledge code for OFFLINE request -. 0 = OFF-LINE Confirmed -. 1 = OFF-LINE Not Allowed -. 2 = Already OFF-LINE -. 3~63 = Reserved

No.	Name	Format	Size	Explanation
34	ONLACK	B	1	Acknowledge code for ONLINE request -. 0 = ON-LINE Confirmed -. 1 = Not allowed to go ON-LINE -. 2 = already ON-LINE -. 3~63 = Hold
35	RCMD	A	-	Remote Command or Command Sequence -. ABORT, CANCEL -. PAUSE, RESUME -. TRANSFER, UPDATE
36	RPTID	U2	2	Report ID
37	SHEAD	B	1	Message header related to the transaction timer
38	SOFTREV	A	Max:6	Software revision code
39	SV	A, B, BOOLEAN, Fx, Ix, K, I, Ux	-	Status Variable Data
40	SVID	U2	2	Status Variable ID
41	SVNAME	A	-	Status Variable Name
42	TEXT	A	Max:80	One line of text
43	TIACK	B	1	Time Acknowledge -. 0 = OK -. 1 = Error, Not Okayed -. 2~63 = Hold
44	TID	U	1	Terminal number -. 0 = single or main terminal -. >0 = additional terminal of the same equipment
45	TIME	A	16	Time of day -. YYYYMMDDhhmmsscc
46	UNITS	A	-	Units identifier

47	V	A, B, BOOLEAN, Fx, lx, K, L, Ux	-	Variable Data
48	VID	U2	2	Variable ID

5.2. Vehicle Variable Data Items

5.2.1. Table-VID Numbering

<i>Variable Type</i>		<i>VID</i>
VIDs	GEM defined variables	1 ~ 50
	SEM defined variables	51 ~ 200
	SFA defined variables	201 ~ 300
Supplier defined VIDs		301 ~

5.2.2. Table-GEM Variables

<i>VID</i>	<i>Name</i>	<i>Description</i>	<i>Class</i>	<i>Format</i>
1	AlarmID	Alarm ID	DVVAL SV	<U4>
2	EstablishCommunicationTimeout	S1F13 Connection timeout	ECV	<U2>
3	AlarmEnabled	Enabled event list	SV	<L [n] 1.<ALID> ... n.<ALID> >

4	AlarmsSet	Active Alarm List	SV	<L [n] 1.<ALID> ... n.<ALID> >
5	Clock	Current Time	SV	A[16]
6	ControlState	Control State -. 1 = offline / equipment offline -. 2 = offline / going to offline -. 3 = offline / host offline -. 4 = offline / local -. 5 = online / remote	SV	<U2>
7	EventsEnabled	Enabled Event List	SV	<L [n] 1. <CEID> ... n. <CEID> >

5.2.3. Table-SEM Variables

VID	Name	Description	Class	Format
51	ActiveCarriers	List current status of all carriers information in the SC database.	SV	<L [n] 1.<CarrierInfo>[] ... n.<CarrierInfo>[] >
52	ActiveTransfers	List of the current status of all carriers in ACTIVE TRANSFER commands.	SV	<L [n] 1.<TransferCommand>[] ... n.<TransferCommand>[] >
53	ActiveVehicles	List of the current status of all vehicles either assigned a TRANSFER command or capable of being assigned a TRANSFER command	SV	<L [n] 1.<VehicleInfo>[] ... n.<VehicleInfo>[] >

212	AlarmText	Alarm Text	DVVAL SV	A[1...128]
401	CarrierCleanStatus	* CST Status = 0 * CST 内 All Mask Clean = 1 * CST 内 1 ↑ Mask Dirty = 2	DVVAL	<U2>
54	CarrierID	Unique ID of the carrier	DVVAL SV	A[1-128]
116	CarrierIDList	The IDs of the Carriers being moved It isn't supported in this SYSTEM if, when this variable is used, it should do CarrierID	DVVAL	<L [n] 1. <A CarrierID> ... n. <A CarrierID> >
55	CarrierInfo	Information regarding events generated by a specific carrier	DVVAL SV	<L [3] <A CarrierID> <A VehicleID> <A CarrierLoc> >
56	CarrierLoc	The carrier's unique location inside the stocker reported by the TSC. -. On Vehicle : VehicleID -. On the Port : PortID	DVVAL SV	A[1-64]

5.2.2. Table-GEM Variables

VID	Name	Description	Class	Format
1	AlarmID	Alarm ID	DVVAL SV	<U4>
2	EstablishCommunicationTimeout	S1F13 Connection timeout	ECV	<U2>
3	AlarmEnabled	Enabled event list	SV	<L [n] 1.<ALID> ... n.<ALID> >
4	AlarmsSet	Active Alarm List	SV	<L [n] 1.<ALID> ... n.<ALID> >

5	Clock	Current Time	SV	A[16]
6	ControlState	Control State -. 1 = offline / equipment offline -. 2 = offline / going to offline -. 3 = offline / host offline -. 4 = offline / local -. 5 = online / remote	SV	<U2>
7	EventsEnabled	Enabled Event List	SV	<L [n] 1. <CEID> ... n. <CEID> >

5.2.3. Table-SEM Variables

VID	Name	Description	Class	Format
51	ActiveCarriers	List current status of all carriers information in the SC database.	SV	<L [n] 1.<CarrierInfo>[] ... n.<CarrierInfo>[] >
52	ActiveTransfers	List of the current status of all carriers in ACTIVE TRANSFER commands.	SV	<L [n] 1.<TransferCommand>[] ... n.<TransferCommand>[] >
53	ActiveVehicles	List of the current status of all vehicles either assigned a TRANSFER command or capable of being assigned a TRANSFER command	SV	<L [n] 1.<VehicleInfo>[] ... n.<VehicleInfo>[] >
212	AlarmText	Alarm Text	DVVA L SV	A[1...128]
401	CarrierCleanStatus	* CST 물류 Status = 0 * In CST, All Mask Clean = 1 * In CST, Above one Mask Dirty = 2	DVVA L	<U2>
54	CarrierID	Unique ID of the carrier	DVVA L SV	A[1-128]
116	CarrierIDList	The IDs of the Carriers being moved It isn't supported in this SYSTEM if, when this variable is used, it should do CarrierID	DVVA L	<L [n] 1. <A CarrierID> ... n. <A CarrierID>

55	CarrierInfo	Information regarding events generated by a specific carrier	DVVA L SV	> <L [3] <A CarrierID> <A VehicleID> <A CarrierLoc> >
56	CarrierLoc	The carrier's unique location inside the stocker reported by the TSC. -. On Vehicle : VehicleID -. On the Port : PortID	DVVA L SV	A[1-64]
VID	Name	Description	Class	Format
58	CommandID	Remote command ID (If a command is generated by the SC, command-ID must begin with the string 'MNL' followed by any arbitrary sequence identified. ex : MANUAL-CommandID format - MNL+"8 Lengths of Equipment Name"+"YYYYMMDDHH24MISS"+"2 Lengths of Sequence"	DVVA L SV	A[1~64]
59	CommandInfo	Command information associated with a particular TRANSFER command	DVVA L SV	<L [3] <A CommandID> <U2 Priority> <U2 Replace> >
57	CommandName	The name of the command sent by Host to the TSC	DVVA L	A[1-120]
93	CommandState	State of Transfer Command -. 0 :Reserve -. 1 :Processing -. 2 :Alternate -. 3 :Complete -. 11 :HOST_Cancel -. 12 :HOST_Abort -. 21 :MNL_Cancel (EQ O/I) -. 22 :MNL_Abort (EQ O/I)	SV	<U2>

		-. 31 :AUTO_Cancel (EQ Auto) -. 32 :AUTO_Abort (EQ Auto) -. 41 :Source_PIOTimeOver -. 42 :Dest_PIOTimeOver -. 43 :Alt_PickUp_PIOTimeOver -. 44 :Alt_Deposit_PIOTimeOver → VHC ≡ 43, 44 NotUse ※ Reference Chapter 10		
74	CommandType	The type of Command being initiated -. 1 : EQ_TRANSFER -. 2 : CANCEL -. 3 : ABORT -. 4 : EQ_AUTO_TRANSFER	DVVAL	A[1-64]

VID	Name	Description	Class	Format
402	CstInfo	CST information	SV	<L [3] <A CarrierID> <U2 CstSize> <A CarrierType> >
373	CarrierType	Carrier Type information	SV	A[1-64]
374	CstPosition	Carrier Position information - 0 : Unknown - 1 : Front - 2 : Rear	SV	<U2>
371	CstInfoList	CST Size List	SV	<L [n] 1. <CstInfo>[] ... n. <CstInfo>[] >

370	CstSize	CST Size Information -. 0 = Unknown -. 1 = Wire -. 2 = Full -. 3 = Quarter -. 4 = Photo-Mask-Cst -. 5 = Even-Mask-Cst -. 6 = Photo-Mask -. 7 = Even-Mask-ONE -. 8 = Even-Mask-QUA -. 9 = Crate -. 10 = Reserved -. 11 = PPBOX -. 12 = Half -. 13 = Even-After-Cst -.14 =FullOHT -.15~20 =Reserved -. 21 = FMM-Mask-Cst -. 22 = TFE-Mask-Cst-4Slot -. 23 = TFE-Mask-Cst-5Slot -. 24 = FMM-Mask -. 25 = TFE-Mask-4Slot -. 26 = TFE-Mask-5Slot -. 31 = Flim-Cst -. 32 = Cell-Cst	DVVAL	<U2>
118	CurrentPortStates	Current State of all the ports	SV	<L [n] 1. <PortInfo>[] ... n. <PortInfo>[] >
60	DestPort	Unique ID of destination port PORT ID	DVVAL SV	A[1-64]

VID	Name	Description	Class	Format
91	EnhancedCarriers	List Current status of all carrier information in the SC database.	SV	<L [n] 1. <EnhancedCarrierInfo>[] ... n. <EnhancedCarrierInfo>[] >

75	EnhancedCarrierInfo	All database information associated with a particular carrier. → <ProcessInfo>[] Delete (since 2014.06.01)	DVVAL SV	<L [7] <A CarrierID> <A VehicleID> <A CarrierLoc> <A InstallTime> <U2 CstSize> <A CarrierType> <U2 CstPosition> >
76	EnhancedTransfers	List current status of all TRANSFER commands.	SV	<L [n] 1. <EnhancedTransferCommand> ... n. <EnhancedTransferCommand> >
205	EnhancedTransferCommand	Information associated with a particular Transfer command. → <ProcessInfo>[] Delete (since 2014.06.01)	DVVAL SV	<L [4] <CommnadInfo>[] <U2 TransferState> <TransferInfo>[] <U2 CarrierCleanStatus> >
119	EnhancedVehicles	List current status of all vehicles available or being used for TRANSFER commands	SV	<L [n] 1. <EnhancedVehicleInfo>[] .. n. <EnhancedVehicleInfo>[] >
403	EnhancedVehicleInfo	Infomation related to a specific vehicle	SV	<L [3] <A VehicleID> <U2 VehicleState> <A VehicleLocation> >

VID	Name	Description	Class	Format
61	EqpName	Unique ID of the SC	ECV	A[1-32]
404	FinalLocation	The Last Destination of Macro Command	DVVAL SV	A[1-64]

405	GlassQuantity	Glass volume in the Carrier	DVVAL	<U4>
210	IDReadStatus	Result Code of an ID read event. -. 0 = Success -. 1 = Failure (BCR NG) -. 2 = Duplicate -. 3 = Mismatch -. 4 = NoCarrier (In the case of no carrier after scanning)	DVVAL	<U2>
204	InstallTime	Time the carrier was created in the SC database.	DVVAL SV	A[16] -. TIME : YYYYMMDDhhmmsscc
406	LotID	ID of Processing Carrier	DVVAL	A[1-64]
407	LotType	Process Type of Processing Carrier -. PP, EE, RR	DVVAL	A[1-64]
408	MainteState	Stocker maint state -. 1=Maintenance -. 2=Not Maintenance	SV	<U2>
252	MonitoredVehicles	List of all vehicle status	SV	<L [n] 1. <MonitoredVehicleInfo>[] ... n. <MonitoredVehicleInfo>[] >

VID	Name	Description	Class	Format
409	MonitoredVehicleInfo	Information for a specific vehicle (Include detailed information)	SV	<L [7] <A VehicleID> <A VehicleCurrentPosition> <A VehicleNextPosition> <U4 AlarmID> <A AlarmText> <U2 VehicleCommunicationState> <U2 MainteState> >
377	MovingPointNo	-. MovingPortNo : Position Vehicle Moving	SV	A[1~64]
115	PortID	Port ID	DVVAL SV	A[1-64]
410	PortInfo	Port information associated with a particular port.	SV	<L [2] <A PortID> <U2 PortTransferState> >
411	PortTransferState	Port Transfer State -. 1 : Outofservice -. 2 : Inservice	SV	<U2>
62	Priority	Remote command priority -. 0 = Not valid -. 1: lowest Priority -. 99 : highest priority	DVVAL SV	<U2>
392	ProcessInfo	Information of Carrier Processing		<L [4] <A LotType> <A LotID> <U4 GlassQuantity> <U4 StepSequence> >
63	Replace	Replace Flag	DVVAL SV	<U2>

VID	Name	Description	Class	Format
64	ResultCode	Execution result of the TRANSFER command -. 0 : success -. 1 : other error -. 4 : ID Read Duplicate -. 5 : ID Read Mismatch -. 6 : ID Read Fail -. 16 : Carrier Size Error -. 64 : Load Error by Vehicle (No Carrier On Vehicle) -. 64 : Unload Error by Vehicle (Carrier On Vehicle) -. 20 : ID Read Duplicate & Carrier Size Error -. 21 : ID Read Mismatch & Carrier Size Error -. 22 : ID Read Fail & Carrier Size Error	DVVAL	<U2>
65	SourcePort	Unique ID of Source port PORT ID	DVVAL SV	A[1-64]
114	SpecVersion	Version of SEMI E88 to which the equipment is compliant.	SV	A[1-20]
412	StepSequence	Processing Step	DVAAL	<U4>
66	TransferCommand	Information associated with a particular TRANSFER command.	DVVAL SV	<L [2] <CommnadInfo>[] <TransferInfo>[] >
79	TransferCompleteInfo	Transfer Complete Info	DVVAL	<L [n] 1. <L [2] <TransferInfo>[] <A CarrierLoc> > ... n. <L [2] <TransferInfo>[] <A CarrierLoc> > >

VID	Name	Description	Class	Format
67	TransferInfo	Carrier information associated with a particular TRANSFER command.	DVVAL SV	<L [6] <A CarrierID> <A SourcePort> <A DestPort> <U2 CstSize> <A FinalLocation> <A CarrierType> >
68	TransferPort	Unique ID of the transfer port	DVVAL	A[1-64]
69	TransferPortList	Transfer report information regarding the arrival and departure of a specific vehicle	DVVAL	<L [n] 1. <A TransferPort> ... n. <A TransferPort> >
202	TransferState	State of Transfer Command -. 1 = queued -. 2 = transferring -. 3 = paused -. 4 = canceling -. 5 = aborting -. 6 = waiting	DVVAL SV	<U2>
73	TSCState	-. 1 = TSC Init -. 2 = paused -. 3 = auto -. 4 = pausing	SV	<U2>
413	VehicleCommunicationState	-. 1 = Communicating -. 2 = Not Communicating ※ If the Error gyeopchyeoteul alarm and communications, ALID in the report because it can not be, communication status is ready to separate status. In this case, the device Error code is reported by ALID	SV	<U2>
414	VehicleControlMode	Control Mode -. 0 = Manual -. 1 = Automatic	SV	<U2>

VID	Name	Description	Class	Format
253	VehicleCurrentDomain	Current Position DOMAIN Name -. 0 = Current station is undefined or Wireless Error	DVVAL SV	A[0-32]
251	VehicleCurrentPosition	Current Position Address -. 0 = Current station is undefined or Wireless Error	DVVAL SV	A[1-10]
256	VehiclePortPosition	Current Destination Port -.0 = Current station is undefined or Wireless Error	DVVAL SV	A[0-32]
415	VehicleDistanceFromCurrentPosition	Distance from current position -. 0 = Current station is undefined or Wireless Error	SV	<U4>
416	VehicleJamState	Traffic Jam State -. 0 = No Jam -. 1 = Jam Exists -. 2 = Stuck	SV	<U2>
417	VehicleNextDomain	Next Position NULL : Current position is undefined (wireless error, UnitID isn't defined in VCL etc.)	SV	A[0-32]
418	VehicleNextPosition	Next Position Address -. 0 = Current station is undefined or Wireless Error	DVVAL SV	A[1-10]
70	VehicleID	The unique ID of vehicle connected to the event	DVVAL SV	A[1-10]
71	VehicleInfo	Information related to a specific vehicle	DVVAL SV	<L [3] <A VehicleID> <U2 VehicleState> <A VehicleLocation> >

419	VehicleOperationState	Operating Status -. 0 = Disconnected -. 1 = Operating -. 2 = Stopped -. 3 = Error -. 4 = Detached (Separated)	SV	<U2>
-----	-----------------------	--	----	------

VID	Name	Description	Class	Format
72	VehicleState	-. 1 = Removed -. 2 = Not assigned -. 3 = Enroute -. 4 = Parked -. 5 = Acquiring -. 6 = Depositing	DVVAL SV	<U2>
117	VehicleLocation	The vehicle's port location The data is only valid if the vehicle is parked, acquiring or depositing	DVVAL	A[0-64]
254	UnitAlarmStatList	UnitAlarmStatList the device to register, is occurring or that a device for communication is abnormal. ※ Includes MTL, ZCU, HID, AD, FD, FFU	SV	<L [n] 1. <UnitAlarmInfo>[] ... n. <UnitAlarmInfo>[] >
420	UnitAlarmInfo	Information for a specific vehicle (Includes detailed information)	SV	<L [6] <A UnitID> <A VehiclePortPosition> <A VehicleNextPosition> <U4 AlarmID> <A AlarmText> <U2 VehicleCommunicationState> >
211	UnitID	The unique ID of the vehicle or Equipment	DVVAL SV	A[1-64]
120	UnitStatusCleable	"Y"	DVVAL	A[1]

421	UpdateCommand		DVVAL	<L [5] <A CommandID> <U2 Priority> <A CarrierID> <A DestPort> <A FinalLocation> >
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5.2.4. Table-SEM Variables [AGV ONLY]

VID	Name	Description	Class	Format
375	ChargeInfo	Information associated with a charging of Vehicle.	SV	<L [2] <A VehicleID> <A ChargePortNo> >
376	ChargePortNo	-. ChargePortNo : Charging Position	SV	A[1~64]

6. COLLECTION EVENT ID

6.1. Control Status Events

CEID	EventName	From Status	To Status	Report ID
1	EquipmentOffline	ON-LINE	OFF-LINE	1
2	ConstrolStatusLocal	OFF-LINE	ON-LINE-LOCAL	1
		ON-LINE-REMOTE		
3	ControlStatusRemote	OFF-LINE	ON-LINE-REMOTE	1
		ON-LINE-LOCAL		

6.2. TSC Status Transition Events

CEID	EventName	From Status	To Status	Report ID
51	AlarmCleared	ALARM	NO ALARM	1
52	AlarmSet	NO ALARM	ALARM	1

53	TSCAutoCompleted	PAUSED	AUTO	1
		PAUSING		
54	TSCAutoInitiated	NONE	TSC INIT	1
55	TSCPauseCompleted	PAUSING	PAUSED	1
56	TSCPPaused	TSC INIT	PAUSED	1
57	TSCPauseInitiated	AUTO	PAUSING	1

6.3. Transfer Command Status Transition Events

<i>CEID</i>	<i>EventName</i>	<i>From Status</i>	<i>To Status</i>	<i>Report ID</i>
101	TransferAbortCompleted	ABORTING	NONE	3
102	TransferAbortFailed	ABORTING	ACTIVE	3
103	TransferAbortInitiated	ACTIVE	ABORTING	3
104	TransferCancelCompleted	CANCELING	NONE	3
105	TransferCancelFailed	CANCELING	QUEUED	3
106	TrnasferCancelInitiated	QUEUED	CANCELING	3
107	TmasferCompleted	ACTIVE	NONE	2
108	TransferInitiated	QUEUED	WAITING	3
109	TransferPaused	TRANSFERRING	PAUSED	2
110	TransferResumed	PAUSED	TRANSFERRING	3
111	Transferring	WAITING or CANCELING	TRANSFERRING	3

6.4. Carrier Status Transition Events

CEID	EventName	From Status	To Status	Report ID
151	CarrierInstallCompleted	NONE	INSTALLED	4
152	CarrierRemoved	INSTALLED	NONE	4
153	VehicleCarrierInstalled	NONE	INSTALLED	16
154	VehicleCarrierRemoved	INSTALLED	NONE	16
251	CarrierIDRead	NONE	NONE	11

6.5. Vehicle Status Transition Events

CEID	EventName	From Status	To Status	Report ID
201	VehicleArrived	ENROUTE	PARKED	5
202	VehicleAcquireStarted	PARKED	ACQUIRING	6
		DEPOSITING		
203	VehicleAcquireCompleted	ACQUIRING	PARKED	6
204	VehicleAssgined	NOT ASSIGNED	ASSIGNED	7
205	VehicleDeparted	PARKED	ENROUTE	5
206	VehicleDepositStarted	PARKED	DEPOSITING	6
		ACQUIRING		
207	VehicleDepositCompleted	DEPOSITING	PARKED	6
208	VehicleInstalled	REMOVED	INSTALLED	8
209	VehicleRemoved	INSTALLED	REMOVED	8
210	VehicleUnassigned	ASSIGNED	NOT ASSIGNED	7
211	VehicleCanNotCallSet	CLEAR	SET	8
212	VehicleCanNotCallClear	SET	CLEAR	8

6.6. Port Status Transition Event

<i>CEID</i>	<i>EventName</i>	<i>From Status</i>	<i>To Status</i>	<i>Report ID</i>
260	PortOutOfService	NONE or IN SERVICE	OUT OF SERVICE	10
261	PortInService	NONE or OUT OF SERVICE	IN SERVICE	10

6.7. Unit Alarm Status Transition Event

<i>CEID</i>	<i>EventName</i>	<i>From Status</i>	<i>To Status</i>	<i>Report ID</i>
503	UnitAlarmCleared	ALARM	NO ALARM	13
504	UnitAlarmSet	NO ALARM	ALARM	13

6.8. UPDATE Command State Transition Event

<i>CEID</i>	<i>EventName</i>	<i>From Status</i>	<i>To Status</i>	<i>Report ID</i>
660	TransferUpdateCompleted	ACTIVE	ACTIVE	3
661	TransferUpdateFailed	ACTIVE	ACTIVE	3
673	DestUpdateCompleted	ACTIVE	ACTIVE	3
674	PriorityUpdateCompleted	ACTIVE	ACTIVE	3

※ TransferUpdateCompleted message was separated into DestUpdateCompleted and PriorityUpdateCompleted.
(for separating Dest and Priority). If both are changed at a time, DestUpdatedCompleted is used.

6.9. Other Event

<i>CEID</i>	<i>EventName</i>	<i>From Status</i>	<i>To Status</i>	<i>Report ID</i>
254	OperationInitiatedAction	NONE	NONE	9
270	MonitoredVehicles	NOT ACTIVE ACTIVE	NOT ACTIVE ACTIVE	14

255	CarrierPreHandoff	TRANSFERRING	TRANSFERRING	6
252	CarrierDestRequest	TRANSFERRING	TRANSFERRING	15
158	CarrierWaitIn	NONE	NONE	33
159	CarrierWaitOut	NONE	NONE	34

6.10. Vehicle Blocking State Transition Event

CEID	EventName	From Status	To Status	Report ID
671	VehicleBlockingStarted	NOT BLOCKED	BLOCKING	32
672	VehicleBlockingEnded	BLOCKING	NOT BLOCKED	32

6.11. Vehicle Charge State Transition Event [AGV ONLY]

CEID	EventName	From Status	To Status	Report ID
662	VehicleChargeSTStarted	NOT CHARGED	START	31
663	VehicleChargedFailed	START CHARGING	FAILED FAILED	31
664	VehicleCharging	START	CHARGING	31
665	VehicleNotCharged	CHARGING FAILED	NOT CHARGED NOT CHARGED	31

7. REPORT ID

7.1. Common ReportID

<i>Report ID</i>	<i>Variable</i>	<i>Report Format</i>
1	EqpName	<L [1] <A EqpName> >

2	EqpName CommandInfo[] TransferCompleteInfo[] ResultCode CarrierCleanStatus CommandState	<pre> <L [6] <A EqpName> <L [3] /* CommandInfo <A CommnadID> <U2 Priority> <U2 Replace> > <L [1] /* TransferCompleteInfo <L [2] <L [5] <A CarrierID> <A SourcePort> <A DestPort> <U2 CstSize> <A FinalLocation> > <A CarrierLoc> > > <U2 ResultCode> <U2 CarierCleanStatus> <U2 CommandState> > </pre>
---	--	--

<i>Report ID</i>	<i>Variable</i>	<i>Report Format</i>
------------------	-----------------	----------------------

3	EqpName CommandInfo[] TransferCompleteInfo[] CarrierCleanStatus CommandState	<pre> <L [5] <A EqpName> <L [3] /* CommandInfo <A CommnadID> <U2 Priority> <U2 Replace> > <L [1] /* TransferCompleteInfo <L [2] <L [5] <A CarrierID> <A SourcePort> <A DestPort> <U2 CstSize> <A FinalLocation> > <A CarrierLoc> > > <U2 CarrierCleanStatus> <U2 CommandState> > </pre>
---	---	---

<i>Report ID</i>	<i>Variable</i>	<i>Report Format</i>
------------------	-----------------	----------------------

4	EqpName CommnadID CarrierID TransferPort CstSize CarrierType	<L [6] <A EqpName> <A CommnadID> <A CarrierID> <A TransferPort> <U2 CstSize> <A CarrierType> >
5	EqpName CommnadID VehicleID TransferPortList	<L [4] <A EqpName> <A CommnadID> <A VehicleID> <L [1] /* TransferPortList <A TransferPort> > >
6	EqpName CommnadID VehicleID CarrierID TrasferPort	<L [5] <A EqpName> <A CommnadID> <A VehicleID> <A CarrierID> <A TransferPort> >
7	EqpName VehicleID CommandID	<L [3] <A EqpName> <A VehicleID> <A CommandID> >

<i>Report ID</i>	<i>Variable</i>	<i>Report Format</i>
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8	EqpName VehicleID	<L [2] <A EqpName> <A VehicleID> >
9	CommandID CommandType CarrierID SourcePort DestPort Priority	<L [6] <A CommandID> <A CommandType> <A CarrierID> <A SourcePort> <A DestPort> <U2 Priority> >
10	EqpName PortID	<L [2] <A EqpName> <A PortID> >
11	EqpName CommandID CarrierID IDReadStatus CarrierLoc	<L [5] <A EqpName> <A CommnadID> <A CarrierID> <U2 IDReadStatus> <A CarrierLoc> >

<i>Report ID</i>	<i>Variable</i>	<i>Report Format</i>
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12	EqpName UnitID AlarmID AlarmText VehicleCurrentDomain UnitStatusCleable	<L [6] <A EqpName> <A UnitID> <U4 AlarmID> <A AlarmText> <A VehicleCurrentDomain> <A UnitStatusCleable> >
13	EqpName UnitID AlarmID AlarmText VehiclePortPosition UnitStatusCleable	<L [6] <A EqpName> <A UnitID> <U4 AlarmID> <A AlarmText> <A VehiclePortPosition> <A UnitStatusCleable> >
14	MonitoredVehicles[]	<L [n] <L [7] <A VehicleID> <A VehicleCurrentPosition> <A VehicleNextPosition> <U4 AlarmID> <A AlarmText> <U2 VehicleCommunicationState> <U2 MainteState> > >
15	EqpName CommandInfo[] TRANSFERCOMPLETEINFO[] VehicleLoc VehicleCurrentPosition	<L 5 <A EQPNAME> <L 3 COMMANDIDINFO <A COMMANDID> <U2 PRIORITY> <U2 REPLACE> > <L 1 TRANSFERCOMPLETEINFO <L 2 <L 5 <A CARRIERID> <A SOURCEPORT> <A DESTPORT>

		<U2 CSTSIZE> <AFINALLOCATION> > <A CARRIERLOC> > > <A VEHICLELOC > <A VEHICLECURRENTPOSITION > >
16	EqpName CommandID CarrierID CarrierLoc VehicleID CstSize CarrierType CstPosition	<L [8] <A EqpName> <A CommnadID> <A CarrierID> <A CarrierLoc> <A VehicleID> <U2 CstSize> <A CarrierType> <U2 CstPosition> >
32	VehicleID CommandID CarrierID MovingPointNo	<L [4] <A VehicleID> <A CommnadID> <A CarrierID> <A MovingPointNo> /* Vehicle Moving Point */ >

7.2. AGV ReportID[AGV ONLY]

Report ID	Variable	Report Format
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31	VehicleID ChargePortNo	<L [2] <A VehicleID> <A ChargePortNo> /* 충전 Station */ >
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7.3. Manual Port ONLY

<i>Report ID</i>	<i>Variable</i>	<i>Report Format</i>
33	EqpName CommandID CarrierID CarrierLoc CstSize	<L [5] <A EqpName> <A CommandID> <A CarrierID> <A CarrierLoc> <U2 CstSize> >
34	EqpName CommandID CarrierID CarrierLoc PortType	<L [5] <A EqpName> <A CommandID> <A CarrierID> <A CarrierLoc> <A PortType> >

8. Remote Command Format

8.1. ABORT Command Format

- This command terminates the activity of a specific TRANSFER command based on CommandID while the command is in the ACTIVE state. This command might not be accepted due to mechanical issues if the stocker is in a specific condition (e.g. moving a carrier) The exact conditions surrounding when the ABORT command is not accepted by the SC must be documented by the stocker SEM equipment supplier.

S2F41

```
<L [2]
  <A 'ABORT'>          /*RCMD*/
  <L [2]
    <L [2]
      <A 'COMMANDID'>    /*CPNAME*/
      <A "CommandID">    /*CPVAL*/
    >
  <L [2]
    <A 'REPLACE'>
    <A "Replace">      * True or False(Global Duplicate abort or normal abort)
  >
>
>
>
```

S2F42

```
<L [2]
  <B 0x04>             /*HCACK*/
  <L [0]
  >
>
```

8.2. CANCEL Command Format

- This command terminates the activity of a specific TRANSFER command based on CommandID while the command is in the QUEUED state. This command must always be accepted by the SC when in the QUEUED state.

S2F41

```
<L [2]
  <A 'CANCEL'>          /*RCMD*/
  <L [1]
    <L [2]
      <A 'COMMANDID'>    /*CPNAME*/
      <A "CommandID">    /*CPVAL*/
    >
  >
>
>
```

S2F42

```
<L [2]
  <B 0x04>             /*HCACK*/
  <L [0]
  >
>
```

8.3. PAUSE Command Format

- This command puts the SC in the PAUSE state.

S2F41

```
<L [2]
  <A 'PAUSE'>          /*RCMD*/
  <L [0]
>
>
```

S2F42

```
<L [2]
  <B 0x04>             /*HCACK*/
  <L [0]
>
>
```

8.4. RESUME Command Format

- This command puts the SC in the AUTO state.

S2F41

```
<L [2]
  <A 'RESUME'>         /*RCMD*/
  <L [0]
>
>
```

S2F42

```
<L [2]
  <B 0x04>             /*HCACK*/
  <L [0]
>
>
```

8.5. SCAN Command Format

- This command is used by Host to scan cassette on the some location using BCR on the OHT.
 - When CarrierID=NULL (Length 0), SCAN CARRIERLOC's Shelf or Port.
 - If there is no such CarrierID, reply with HCACK 6.
- When this is no carrier on CarrierLoc, reply with Parameter Invalid (HCACK3).

S2F41

```
<L [2]
  <A 'SCAN'>           /*RCMD*/
  <L [3]
    <L [2]
      <A 'COMMANDID'>  /*CPNAME*/
      <A "CommandID">  /*CPVAL*/
    >
    <L [2]
      <A 'CARRIERID'> /*CPNAME*/
      <A "CarrierID">   /*CPVAL*/
  >
>
```

```

    <L [2]
    <A 'CARRIERLOC'>      /*CPNAME*/
    <A "CarrierLoc">        /*CPVAL*/
  >
>
>

S2F42
<L [2]
  <B 0x04>                /*HCACK*/
  <L [0]
  >
>

```

8.6. UPDATE Command Format

- This command is used by the Host to update previous command the SC excuting.
- In the case that Priority is NULL (Length 0), Keep the priority. : In the case that only DestPort is changed
- In the case that DestPort is NULL (Length 0), Keep the DestPort. : In the case that only Priority is changed

S2F41

```

<L [2]
  1. <A "RCMD" >          * "UPDATE"
  2. <L [5]
    1. <L [2]
      1. <A 'COMMANDID' >
      2. <A "CommandId" >
    >
    2. <L [2]
      1. <A 'CARRIERID' >
      2. <A "CarrierID" >
    >
    3. <L [2]
      1. <A 'PRIORTY' >
      2. <U2 "Priority" >
    >
    4. <L [2]
      1. <A 'DESTPORT' >
      2. <A "DestPort" >
    >
    5. <L [2]
      1. <A 'FINALLOCATION' >
      2. <A "FinalLocation" >
  >
>

```

S2F42

```

<L [2]

```

```

<B 0x04>          /*HCACK*/
<L [0]
>
>

```

8.7. TRANSFER Command Format

S2F49

```

<L [4]
  <U4 0 >          * DATAID "0 : fixed"
  <A[40] OBJSPEC >      * OBJSPEC "Not Used"
  <A[32] 'TRANSFER' >  * RCMD
  <L [2]
    <L [2]
      <A[32] 'COMMANDINFO' >  * CPNAME
      <L [3]
        <L [2]
          <A 'COMMANDID' >
          <A "CommandID" >
        >
        <L [2]
          <A 'PRIORITY' >
          <U2 "Priority" >
        >
        <L [2]
          <A 'REPLACE' >
          <U2 "Replace" >
        >
      >
    >
  >
  <L [2]
    <A[32] 'TRANSFERINFO' >  * CPNAME
    <L [6]
      <L [2]
        <A 'CARRIERID' >
        <A "CarrierID" >
      >
      <L [2]
        <A 'SOURCEPORT' >
        <A "SourcePort" >      * PORT ID
      >
      <L [2]
        <A 'DESTPORT' >
        <A "DestPort" >      * PORT ID
    >
  >

```



```

>
<L [2]
  <A[13] 'CARRIERTYPE' >
  <A[11] 'CarrierType' >
>
>
>

```

8.9. REMOVE Command Format

[REMOVE]

MSG:S2F41[Host Command Send : Remove]

```

<L [2]
  <A[6] 'REMOVE' >
  <L [1]
    <L [2]
      <A[9] 'CARRIERID' >
      <A[8] '1FS9063E'>
    >
  >
>

```

9. DESCRIPTION

9.1. The Reason of CommandState

<i>CEID</i>	<i>EventName</i>	<i>Result Code</i>	<i>Command State</i>	<i>Command</i>	<i>Status</i>
101	TransferAbortCompleted	-	12	HostCommand	Abort
			22	EQ O/I	Abort
			32	EQ Auto	Abort
102	TransferAbortFailed	-	0	HostCommand EQ O/I	Reserve

			1	EQ Auto	Processing
			2		Alternate
103	TransferAbortInitiated	–	0	HostCommand EQ O/I EQ Auto	Reserve
			1		Processing
			2		Alternate
104	TransferCancelCompleted	–	11	HostCommand	Cancel
			21	EQ O/I	Cancel
			31	EQ Auto	Cancel
105	TransferCancelFailed	–	0	HostCommand EQ O/I EQ Auto	Reserve
			1		Processing
			2		Alternate

CEID	EventName	Result Code	Command State	Command	Status
106	TrnasferCancelInitiated	–	0	HostCommand EQ O/I EQ Auto	Reserve
			1		Processing
			2		Alternate
107	TrnasferCompleted	0,1,4,5,6,16, 20,21,22	3	HostCommand EQ O/I EQ Auto	Complete
		64	41		Source_PIOTimeOver
			42		Dest_PIOTimeOver

108	TransferInitiated	–	0	HostCommand EQ O/I EQ Auto	Reserve
109	TransferPaused	–	0	HostCommand EQ O/I EQ Auto	Reserve
			1		Processing
			2		Alternate
110	TransferResumed	–	0	HostCommand EQ O/I EQ Auto	Reserve
			1		Processing
			2		Alternate
111	Transferring	–	1	HostCommand EQ O/I EQ Auto	Processing

<i>CEID</i>	<i>EventName</i>	<i>Result Code</i>	<i>Command State</i>	<i>Command</i>	<i>Status</i>
661	TransferUpdateFailed	–	0	HostCommand EQ O/I EQ Auto	Reserve
			1		Processing
			2		Alternate
673	DestUpdateCompleted	–	0	HostCommand EQ O/I EQ Auto	Reserve
			1		Processing

			2		Alternate
674	PriorityUpdateCompleted	-	0	HostCommand EQ O/I EQ Auto	Reserve
			1		Processing
			2		Alternate